

EEE 359

# Engineering Project Management

# Lecture 05

# Project Management Plan

# Project Management Plan

The project management plan is the document that describes how the project will be executed, monitored and controlled, and closed. It integrates and consolidates all of the subsidiary management plans and baselines, and other information necessary to manage the project. The needs of the project determine which components of the project management plan are needed.

Project management plan (as per Project Management Institute) components include but are not limited to:

- Subsidiary Management Plan
- Baselines
- Additional Components

# Subsidiary Management Plan

- **Scope management plan:** Establishes how the scope will be defined, developed, monitored, controlled, and validated
- **Requirements management plan:** Establishes how the requirements will be analyzed, documented, and managed
- **Schedule management plan:** Establishes the criteria and the activities for developing, monitoring, and controlling the schedule
- **Cost management plan:** Establishes how the costs will be planned, structured, and controlled
- **Quality management plan:** Establishes how an organization's quality policies, methodologies, and standards will be implemented in the project
- **Resource management plan:** Provides guidance on how project resources should be categorized, allocated, managed, and released

# Subsidiary Management Plan

- **Communications management plan:** Establishes how, when, and by whom information about the project will be administered and disseminated
- **Risk management plan:** Establishes how the risk management activities will be structured and performed
- **Procurement management plan:** Establishes how the project team will acquire goods and services from outside of the performing organization
- **Stakeholder engagement plan:** Establishes how stakeholders will be engaged in project decisions and execution, according to their needs, interests, and impact

# Baselines

- **Scope baseline:** The approved version of a scope statement, work breakdown structure (WBS), and its associated WBS dictionary, which is used as a basis for comparison
- **Schedule baseline:** The approved version of the schedule model that is used as a basis for comparison to the actual results
- **Cost baseline:** The approved version of the time-phased project budget that is used as a basis for comparison to the actual results

# Additional Components

Most components of the project management plan are produced as outputs from other processes, though some are produced during this process. Those components developed as part of this process will be dependent on the project; however, they often include but are not limited to:

- **Change management plan:** Describes how the change requests throughout the project will be formally authorized and incorporated
- **Configuration management plan:** Describes how the information about the items of the project (and which items) will be recorded and updated so that the product, service, or result of the project remains consistent and/or operative

# Additional Components

- **Performance measurement baseline:** An integrated scope-schedule-cost plan for the project work against which project execution is compared to measure and manage performance
- **Project life cycle:** Describes the series of phases that a project passes through from its initiation to its closure
- **Development approach:** Describes the product, service, or result development approach, such as predictive, iterative, agile, or a hybrid model
- **Management reviews:** Identifies the points in the project when the project manager and relevant stakeholders will review the project progress to determine if performance is as expected, or if preventive or corrective actions are necessary

# Project Model Plan (British Standard Institute)

\*No need to memorize!\*

## **General**

1. Foreword
2. Contents, distribution and amendment record
3. Introduction
  - 3.1 Project diary
  - 3.2 Project history

## **The Why**

4. Project aims and objectives
  - 4.1 Business case

## **The What**

5. General description
  - 5.1 Scope
  - 5.2 Project requirement
  - 5.3 Project security and privacy
  - 5.4 Project management philosophy
  - 5.5 Management reporting system

# Project Model Plan (British Standard Institute)

\*No need to memorize!\*

## **The When**

6. Program management

6.1 Program method

6.2 Program software

6.3 Project life cycle

6.4 Key dates

6.5 Milestones and milestone slip chart

6.6 Bar chart and network if available

## **The Who**

7. Project organization

8. Project resource management

9. Project team organization

9.1 Project staff directory

9.2 Organizational chart

9.3 Terms of reference (TOR)

(a) For staff

(b) For the project manager

(c) For the committees and working group

# Project Model Plan (British Standard Institute)

\*No need to memorize!\*

## **The Where**

- 10. Delivery requirements
  - 10.1 Site requirements and conditions
  - 10.2 Shipping requirements
  - 10.3 Major restrictions

## **The How**

- 11. Project approvals required and authorization limits
- 12. Project harmonization
- 13. Project implementation strategy
  - 13.1 Implementation plans
  - 13.2 System integration
  - 13.3 Completed project work
- 14. Acceptance procedure
- 15. Procurement strategy
  - 15.1 Cultural and environmental restraints
  - 15.2 Political restraints
- 16. Contract management

# Project Model Plan (British Standard Institute)

\*No need to memorize!\*

- 17. Communications management
- 18. Configuration management
  - 18.1 Configuration control requirements
  - 18.2 Configuration management system
- 19. Financial management
- 20. Risk management
  - 20.1 Major perceived risks
- 21. Technical management
- 22. Tests and evaluations
  - 22.1 Warranties and guarantees
- 23. Reliability management
  - 23.1 Availability, reliability and maintainability (ARM)
  - 23.2 Quality management
- 24. Health and safety management
- 25. Environmental issues
- 26. Integrated logistic support (ILS) management
- 27. Close-out procedure

# Work Breakdown Structures (WBS)

# Work Breakdown Structure

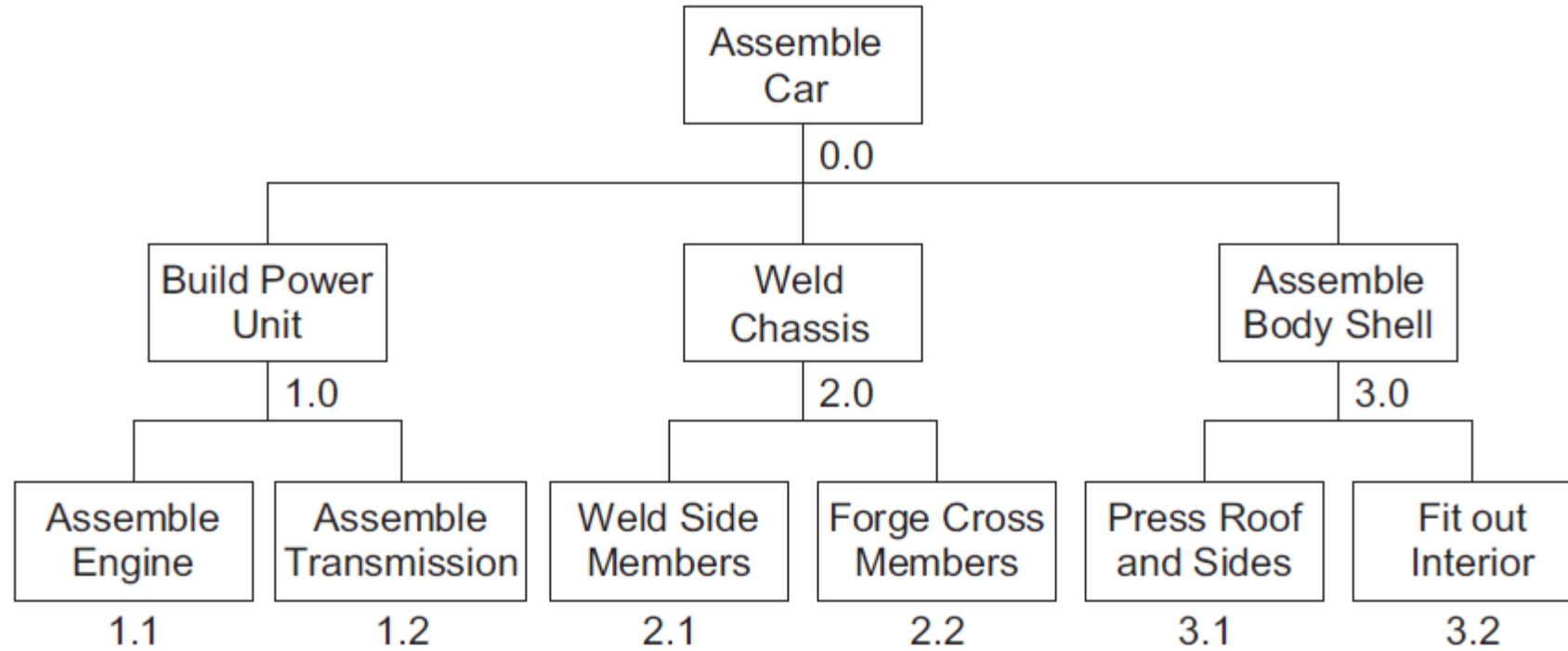
The WBS is a **hierarchical** decomposition of the total work to be carried out by the project team to accomplish the project objectives and create the required deliverables. Each descending level of the WBS represents an increasingly detailed definition of the project work.

Decomposition is a technique used for dividing and subdividing the project scope and project deliverables into smaller, more manageable parts. The level of decomposition is often guided by the degree of control needed to effectively manage the project.

Decomposition of the total project work generally involves the following activities:

- Identifying and analyzing the deliverables and related work
- Structuring and organizing the WBS
- Decomposing the upper WBS levels into lower-level detailed components
- Developing and assigning identification codes to the WBS components
- Verifying that the degree of decomposition of the deliverables is appropriate

# Work Breakdown Structure



Work breakdown structure

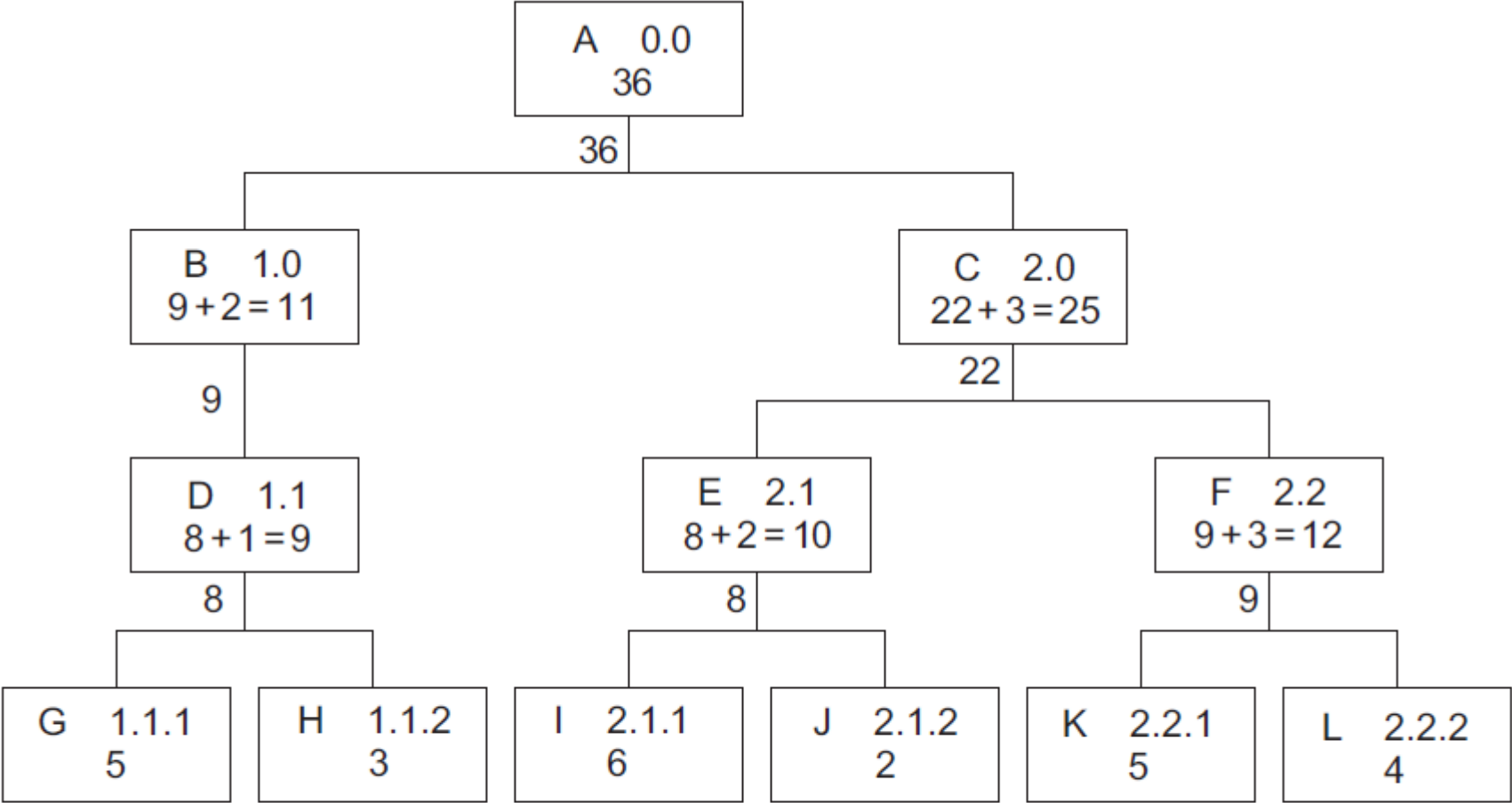
# Cost Breakdown Structure

A Cost Breakdown Structure (CBS) is a **hierarchical** representation of all project costs grouped by categories or components.

It shows how the total project cost is **distributed** among different activities, resources, and deliverables.

It helps project managers **estimate, track, and control** the project budget.

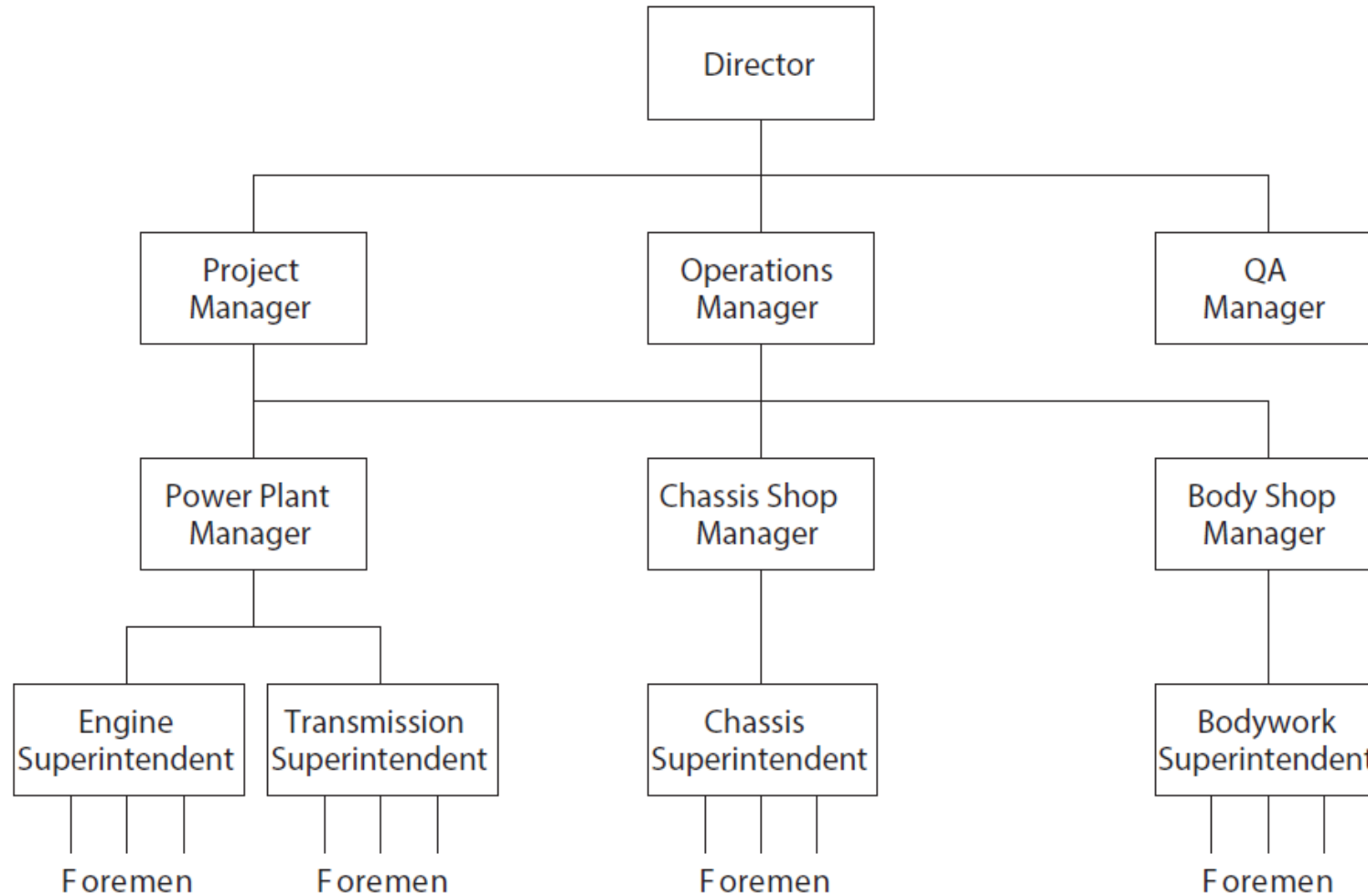
# Cost Breakdown Structure



Cost breakdown structure.

# Organization Breakdown Structure

An Organization Breakdown Structure (OBS) is a project management tool that shows how the project team is organized and who is responsible for each part of the project.



Organization breakdown structure.

# Responsibility Assignment Matrix

A RAM shows the project resources assigned to each work package. It is used to illustrate the connections between work packages, or activities, and project team members. On larger projects, RAMs can be developed at various levels.

RAM links project tasks (from WBS) with people or roles (from OBS) to clearly show who is responsible for what.

The matrix format shows **all activities associated with one person** and **all people associated with one activity**. This also ensures that there is **only one person accountable** for any one task to avoid confusion about who is ultimately in charge or has authority for the work.

# Responsibility Assignment Matrix

The sample chart shows the work to be done in the left column as activities. The assigned resources can be shown as individuals or groups. The project manager can select other options, such as “lead” and “resource” designations, as appropriate for the project. A **RACI** chart is a useful tool to use to ensure clear assignment of roles and responsibilities when the team consists of internal and external resources.

| RACI Chart                                             | Person |     |        |      |    |
|--------------------------------------------------------|--------|-----|--------|------|----|
| Activity                                               | Ann    | Ben | Carlos | Dina | Ed |
| Create charter                                         | A      | R   | I      | I    | I  |
| Collect requirements                                   | I      | A   | R      | C    | C  |
| Submit change request                                  | I      | A   | R      | R    | C  |
| Develop test plan                                      | A      | C   | I      | I    | R  |
| R = Responsible A = Accountable C = Consult I = Inform |        |     |        |      |    |

**R** = Responsible → Person who does the work (atleast one)  
**A** = Accountable → Person ultimately answerable (only one)  
**C** = Consulted → People who provide input (optional)  
**I** = Informed → People kept updated

Sample RACI Chart

Thanks